

CO4 Imaging for the Life Sciences

Summary

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1. Introduction:

You can describe the color perception of the human eye

- Color perception in the human eye works with photo receptor cells called cones. There are three different types for the colors (different ranges of wavelengths) green, blue and red. The blue cones are underrepresented. Signals from these cones are then processed by the brain.

You have a basic understanding of the Nyquist-Shannon sampling theorem

- To reconstruct a signal completely it must be sampled with a frequency f_s that is twice as large as the highest frequency f_{max} in the signal. $f_s > 2 * f_{max}$. If f_s is too small, aliasing is happening. This means higher frequencies are aliased by wrong lower frequencies, which results in a distortion. In image processing the spatial frequency which has to be sampled is determined by the granularity of the image. Patterns that change from light to dark can cause distortions if they change faster (are small in the image) than the sampling frequency. This leads to Aliasing effects, such as Moiré-Patterns, wrong structures (checked shirts) and others.

You know what digitization means

- Digitization is the transformation of an analog signal into a digital computer readable signals. Digitization is done by sampling the analog signal with given intervals and quantization. Which results in a finite set of levels.

You know the advantages of digital images

- Storage, sharing, reproduction, editing flexibility, instant viewing, cost effectiveness, versatility (easy to change format), metadata integration, remote access, enhanced preservations, less noise

You can give some examples of direct and indirect imaging methods

- Direct:
 - Photography, Microscope, Telescope
- Indirect:
 - Particle Waves (electron microscopy)
 - Electromagnetic Waves (MRI, X-Ray)
 - Pressure Waves (Medical Ultrasound)

2. Optical Imaging

You can name the most important optical imaging methods used in preclinical biomedical research

- Fluorescence microscopy, bioluminescence imaging, confocal microscopy, multiphoton microscopy

Within a cell, you can point out the specific structures and processes which can be imaged using molecular (optical) imaging techniques

- Cell membrane/wall, nucleotide, organelles, proteins, RNA/DNA, signal pathways

You can explain how and for what purpose optical imaging is employed in drug development

- Pathways / API (active pharmaceutical ingredient) distribution, effects / mechanisms, toxicity

You can compare the different optical imaging methods discussed regarding spatial resolution and sensitivity

- Fluorescence microscopy:
- Bioluminescence imaging: low resolution, high sensitivity
- Confocal Microscopy: high resolution, mediocre sensitivity
- Multiphoton Microscopy: high resolution, high penetration depth

You can explain how fluorescent signals are generated

- A fluorescent signal can be generated through excitation of a fluorescent molecule / protein with light in a suitable wavenumber. The light excites electrons which jump from a low energy level to a higher one. When they jump back, they emit the energy in the form of light.

You can explain why fluorescent probes used for in vivo imaging should have properties in the NIR range of the spectrum

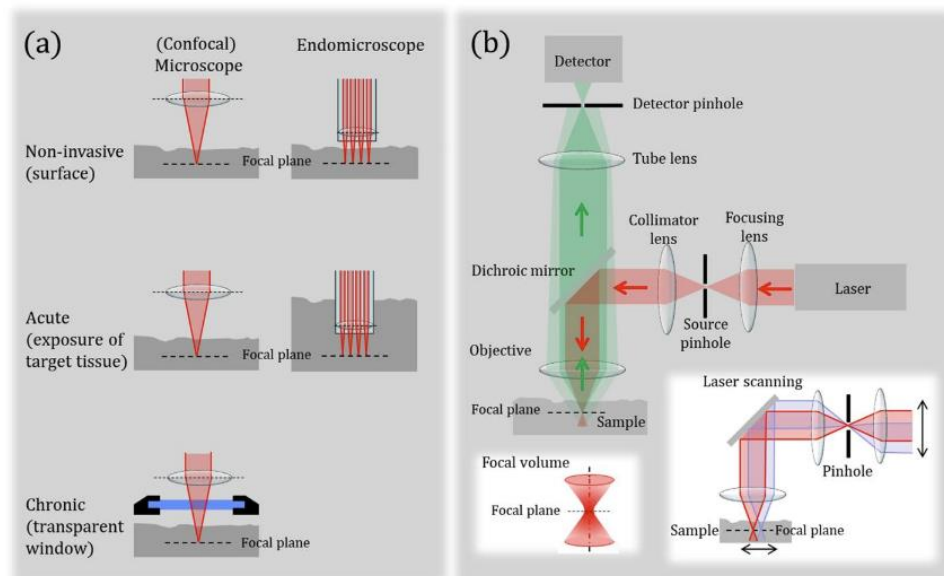
- NIR has low scattering, tissue absorbing, auto fluorescence and high penetration depth. One reason is that NIR is less absorbed by water and hemoglobin

You can give examples for fluorescent probes (low molecular weight probes as well as fluorescent proteins), which have their excitation/emission spectra in the NIR range and are typically used for in vivo imaging purposes

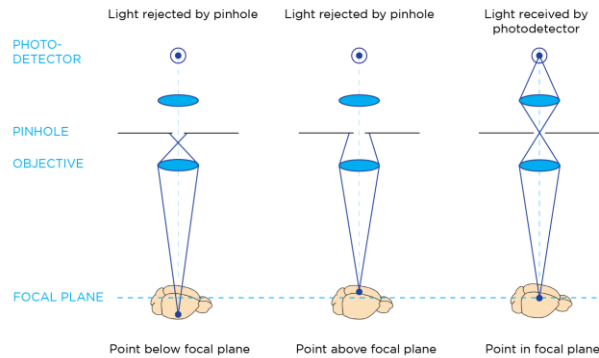
- Fluorescent dyes / probes (Indocyanine dyes: Cy5, Cy5.5, Cy7, DBCy7)
- Fluorescent proteins (GFP, RFP, StayGold, myCherry)

You can explain the principles underlying confocal and multiphoton microscopy

- Confocal Microscopy:
 - Optical imaging technique, with increased resolution and contrast. Accomplished by an illuminate and excite only points close to the confocal plane (instead of the whole sample). A pinhole in front of the detector allows to eliminate out-of-focus signals.

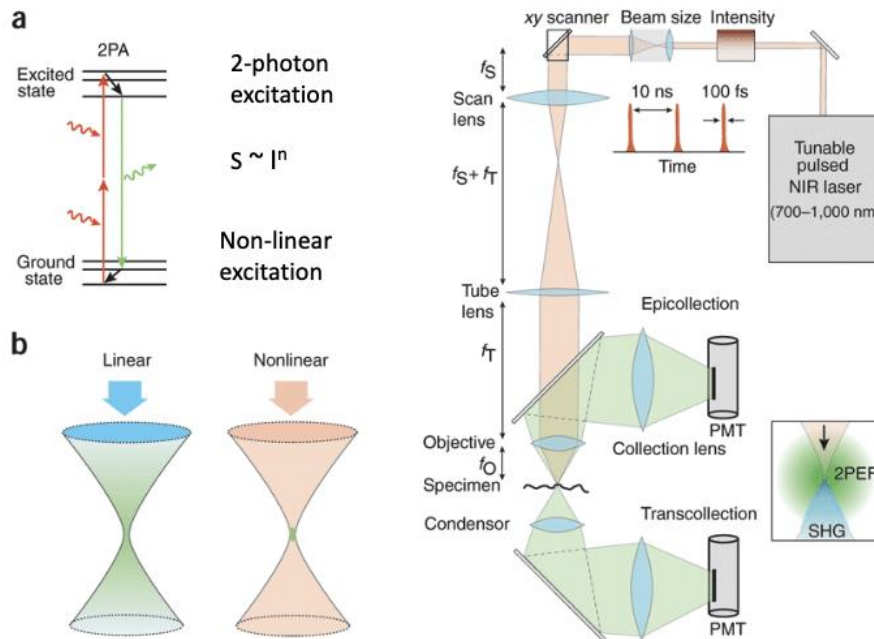


Of



- **Multiphoton Microscopy**

- Optical imaging technique, where the image is sequentially produced by scanning across the sample with the laser. Instead of excitations with higher energy wavelengths and emission with slightly lower energy, in multiphoton microscopy the sample is excited with two or multiple photons of lower energy at once to emit higher energy wavelengths. This approach results mostly in excited fluorophores in the focus of the laser → high spatial resolution
- Usually the excitation is done with NIR light



You can provide examples for applications where such intravital microscopy techniques are the appropriate imaging techniques used

- Real time imaging of live tissue
- Observations of tumors in microenvironments
- Blood flow
- Immune cell interactions
- Trafficking of dendritic cells through lymphatics
- ...

You can explain how bioluminescence imaging works and for which type of experiments bioluminescence imaging is used

- Bioluminescence in general

- In bioluminescence no external excitation is needed to initiate fluorescence. Excitation is achieved through a chemical reaction. $A + B \rightarrow C^* \rightarrow C + h\nu$: A and B react and form an already excited Product C^* , through emission of light it can stabilize itself. An example of such a molecule is Luciferase
- Bioluminescence imaging
 - Is a cheap readout imaging technology in pharmaceutical development
 - Can only be used in preclinical tests, since genetically modified cells / proteins are used
- Applications / Type of Experiments
 - Monitoring immune cell activation in tumor model
 - Genetically modified cancer cells (produce Luciferase)
 - This allows to measure tumor growth respectively the blockage if a potential API against cancer is added

What is hypoxic cell signalling and how does it work?

- Hypoxic Inducible Factor
 - HIF is a transcription factor to produce protein related to Oxygen-to-tissue delivery
 - If HIF is hydroxylated (if O₂ concentration is normal) this leads to its degradation through proteasome
 - If the HIF is NOT hydroxylated (which occurs if O₂ concentration is low) it binds to a HIF-beta unit and then to DNA which leads to the transcription of genes
 - Genes: Angiogenesis, EPO generation, etc.
- Hypoxic cell signalling
 - Since low oxygen levels correlate to tumour growth it is of interest to visualize this
 - This is done through the modification of the HIF-1 alpha gen, a firefly luciferin gen is attached which induces visibility whenever HIF is stable
 - Another possibility is to attach the gen of the firefly luciferin next to the genes promoted through HIF, so it is produced if the promotor is activated

3. MRI:

where does the MRI signal come from?

- The signal is produced by the changes in a magnetic field. The changes are caused by Hydrogen atoms which are going back to a relaxed state after they were aligned in the magnetic field

What are the main components of an MRI system?

- Electro-magnets in vacuum containers
- Superconducting solenoid coils
 - Magnet Coils
 - Gradient Coils
 - RF Coils (Receiving)
- Cryocooler
- Liquid Helium
- Patient table
- A/D System
- Processing Units (computers)

How is spatial resolution realized in MRI?

- The spatial resolution can be realized with varying gradient fields. This allows to localize a signal in three dimensions

What does Larmor frequency means?

- The Larmor frequency ω_0 (MHz) is the frequency with which the direction of the magnetic moment (axis of directed molecule) spins around the direction of the magnetic field B_0 (Tesla, A/m). The Frequency is proportional to the magnetic field, scaled by a factor γ (MHz / T)

$$\omega_0 = \gamma * B_0$$

What needs to be considered to operate an MRI safely?

- Magnetic field
 - Magnetic forces / torques
 - Implants
 - Ferromagnetic objects
- Radio Frequency
 - Heating effects
 - Interference with implants
- Gradient Field
 - Nerve Stimulations
 - Acoustic noise
- Patient handling
 - Communication / Controlling
 - Claustrophobia
- → avoid metallic objects, monitor patients, protection against strong magnetic field, RF, Gradient

What is relaxation?

- Something I do not experience right now since there are exams ahead...
- The T1 and T2 describes how the relaxations of excited hydrogen atoms proceed after a pulse
- Longitudinal Relaxation (T1)
 - Change in spin direction of the proton
 - Umbrella shaped
 - Fat is bright
- Transverse relaxation (T2)
 - Spinning of protons are in phase (synchronized)
 - Fan shaped
 - Is dominant and appears first
 - Water is bright

What is a gradient?

- Varying magnetic field in a particular direction
- This means that the Hydrogen atom spin in different frequencies in different locations
- Makes the RF pulse selective to a specific slice of tissue
- Allows to collect data from a specific area
- Can spatially localize the MR-Signal → where does the signal come from?

What is a Sequence and which are the most important MR sequences?

- Sequences:
 - Precise RF-Pulses and gradients to generate MR signal to detect and img creation
 - Depending on the weighting of T1 (fat) and T2 (water) different tissue contrasts can be received
 - TE Time to Echo, TR time to repetition (of a RF Pulse)
 - Short TR: emphasize T1, for example high liquor brain tissue contrast

- Long TR increases contrast between white and gray matter
- Long TE: emphasize T2
- Different TR and TE combinations makes it possible to contrast different tissues
- Short TR and long TE results in bad resolution and bad contrast
- Gradient Echo Sequence / Multi-Gradient echo sequence
 - Simple pulse sequence with one 90° RF Pulse (NO 180° Pulse) and gradient application afterwards
 - Short TE
 - Short repetition time TR
 - T1 Weighting
 - Angiography (only one pulse) (Vascular systems, Blood and Lymphatics)
- EPI
 - Echo Planar Imaging (extreme fast method)
 - Applied when small movements interfere
 - If long echo time is necessary (fMRI)
 - If distortions and poor resolutions are acceptable
 - One RF pulse with multiple gradients afterwards
- Spin-Echo
 - Rarely used (T1 weighting brain/spine)
 - 90° RF Pulse followed by a 180° RF Pulse
 - 90° Pulse → precess in xy plane
 - 180° Pulse → rephasing / refocusing pulse, brings H+ back into phase
 - 180° Pulse change spin direction
 - Spin echo is used to reduce magnetic field inhomogeneities
- TSE
 - Turbo Spin Echo (multiple 180° pulses)
 - Workhorse from head to toe
 - T2 weighting / spin density weighting
 - Reduced measuring times,

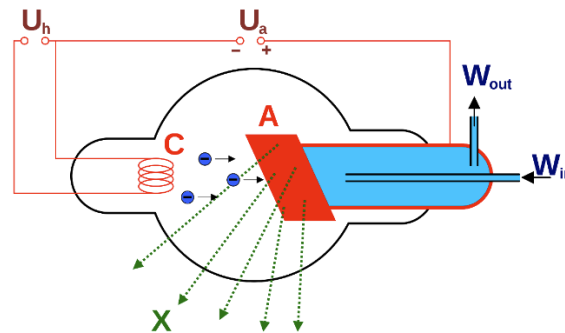
Sequence	Characteristic	Advantages	Disadvantages	Applications
GRE	No 180° pulse Single 90° pulse T2*	Fast T2* sensitive	No T2 weighting Vulnerable to artifacts	fMRT 3D sequences
EPI	Extreme fast k-space in one shot	Extreme fast Insensitive to movement	Low resolution Distortions	fMRT Diffusion/perfusion Emergency
SE	90° +180° pulse Robust against inhomogeneities	High img quality Insensitive to artifacts Good tissue characterization	Long acquisition time Bad for 3d img	Standard seq. Rarely used (T1, T2, etc.)
TSE/FSE	90° + n*180° pulse Faster than SE	High quality Fast acquisition Versatile	Artifacts (echo trains) Blurring	Work horse T2 (Joints, Spine, Abdomen, Brain)

4. X-Ray, CT, PET, SPECT

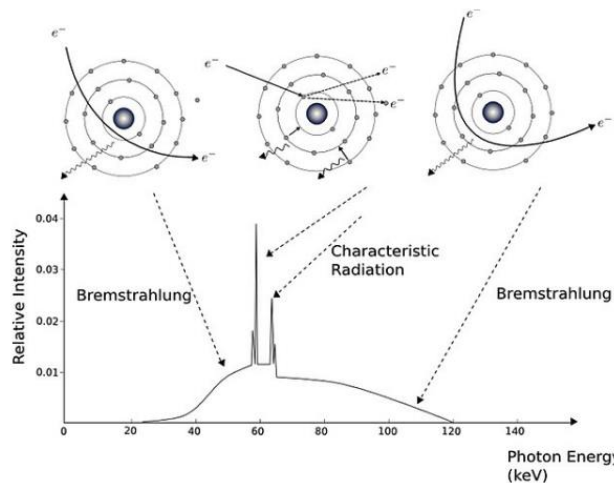
You can explain how X-Rays are generated incl. the two types of radiation

- X-Ray's are generated with a X-Ray tube
 - The cathode C (a heating coil) is heated by the Voltage U_h which leads to electron emission

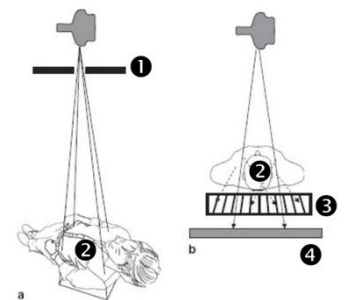
- The electrons travel through a vacuum chamber to the Anode. The Voltage U_a provides the electric field for the attraction
- The impact of the electrons on A lead to emission of X-Rays X
- The Anode A is cooled with water



- Radiation types
 - Two types of radiation are emitted
 - Bremsstrahlung (continuous line, flying electrons, slowed down by attractive force of positive nucleus)
 - Characteristic Radiation (peaks, flying electrons which hit electrons in shells of the anode)



- 2D projection (No Tomography)
 - 1. Collimator (cover that narrows beam to prevent unnecessarily radiation of parts outside target)
 - 3. Target
 - 3. Anti-scatter grid (Suppresses Compton scattered X-Rays)
 - 4. Detector (solid-state detectors)



You can explain the term “tomography”

- Imaging technique which creates cross sections of an object/body, by combining (multiple) projections taken from different angles

You know what a Sinogram is and how a basic back-projection (“smear-back”) works

- Sinogram
 - Includes all data to reconstruct a cross-section image
 - It represents the data from all angles taken
 - Matrix, columns correspond to a Radon transformation R_θ
 - Number of cols = applied rotation angles θ
 - Rows = resolution of CT detector

- Smear back projection
 - To reconstruct a cross-section from a sinogram
 - The measured data (intensity) are projected back to construct an image
 - If this is done without corrections, the image is slightly blurry or smeared
 - With normalization and filters the image can be reconstructed with increased sharpness

You know what Hounsfield Units are and you can explain what so-called “Windowing” is

- Housefield Units
 - Housefield-scale shows how much a tissue can lessen the xray signal
 - The higher the linear attenuation coefficient μ of a tissue is the higher its HU is
+1000 = bone, -1000 = air, 0 = water
 - $HU = 1000 * \frac{\mu - \mu_{water}}{\mu_{water} - \mu_{air}}$ (Linear transformation of attenuation coefficient to HU)
- Windowing
 - The measured CT values (attenuations coefficient) can be transformed to HU, these can then be mapped to grayscale
 - This allows to highlight or dim certain types of tissue
 - Example: lung window (-1450 – 250 HU), sub tissue window (-250 – 150 HU), bone window (-250 – 2250 HU)

You can explain the basic principles of nuclear imaging (PET/SPECT)

- In nuclear imaging a tiny amount of radioactive material bound to a tracer (radiotracer) is injected
- The molecule is distributed in the body depending on the tracer (Glucose → heart, brain, tumors etc.)
- This tracer emits gamma-rays which are scattered by the body and exit it in all directions
- The collimator accepts only photons in a certain direction (filter/lens, sharpens)
- The gamma-rays are then converted to visible photons with a scintillation crystal
- The visible light is detected with photomultiplier tubes
- PET and SPECT aim to understand organ function (instead of organ anatomy)

You can explain the difference between PET and SPECT

- PET = Positron emission tomography
 - Positron emitting radioisotopes (Fluorine)
 - Positron hits electron, resulting in two gamma-rays travelling in opposite directions
 - Usually, tubular device with ring shaped detectors around the patient
- SPECT = Single photon emission tomography
 - Gamma emitting radioisotopes (Technetium)
 - Technetium only produces a single gamma ray which is detected
 - Two rectangular detectors that rotate around the patient
- Combination with CT
 - PET and SPECT are often combined with CT scans
 - This allows to combine metabolic and anatomic information

5. Optoacoustic Imaging

Why do we need optoacoustic imaging?

- Noninvasive method with much higher spatial resolution than conventional optical imaging
- Combination of light and ultrasound provides optical contrast with higher penetration depth
- Five-dimensional imaging is possible (3D, time, optical wavelength)

- In conclusion: optoacoustic imaging provides high resolution with high penetration depth which overcomes limitations of common optical imaging

How optoacoustic imaging is done?

- Signal generation: short, pulsed laser pulses (<100 ns, usually ~10 ns)
- The more energy of the pulse is absorbed by the tissue the larger the pressure rise (proportional relation)
- The pressure rise in the tissue results in an ultrasound wave which can then be detected and transformed into an image

How optoacoustic facilitates biological discovery?

- Optoacoustic provides high resolution images with high penetration depth in real time
- This allows to get new insights in tissue structures, blood vessels and molecular compositions
- Which supports research of disease mechanisms and treatment effects

Can we use optoacoustic for medical diagnostics?

- Tumor detection, monitoring blood oxygenation, assessing tissue perfusion
- Noninvasive, high information imaging modalities
- Clinical Trials: Breast Cancer, Skin Cancer, Inflammatory Bowel Disease, Vascular Diseases, Atherosclerosis, Thyroid Cancer, Lymphography, Muscular Dystrophy

What are M, AB & C modes in ultrasound measurements?

- M: motion (signal as function of time)
- AB: Amplitude, one single signal to B-mode (brightness) (
- C: Contrast (scans in x and y directions)

What is echogenicity?

- These are different behaviors of objects/tissue concerning how they reflect sound compared to their surrounding
- Isoechoic: same sound reflection as surrounding
- Hyperechoic: more sound reflection as surrounding
- Hypoechoic: less sound reflection as surrounding
- Anechoic: no internal echoes (tissue filled with water, does not reflect sound)

What is important to know about the resolution of Ultrasound devices?

- The resolution depends on the shape of the array
- The axial resolution (along beam) is always greater than the lateral (perpendicular to beam)
- The axial resolution is frequency dependent, the lateral depends on the transducer array shape
- The higher the frequency the narrower the beam

Which bodyparts/things can be distinguished with different frequencies/wavelengths?

- Oxyhemoglobin (red)
- Desoxyhemoglobin (blue)
- Melanin
- Lipids/Water

Summarize / Overview Table

How does xy work?

What are common processes?

What are advantages, disadvantages, limitations?

What is input, output what is actually recorded?
How is the recorded signal transferred into the image?